

Countermeasures to Receiver-end Spoofing

Method	Test statistic	Function	Limitation
Method 1	Absolute signal power	Limit the spoof signal power	Antenna attitude and environment related
Method 2	Signal power changing rate	Detect stationary spoof station	Antenna attitude and environment related
Method 3	Relative signal strengths on all carriers	Detect spoofing on single carrier	Affected by ionosphere refraction
Method 4	Range rate	Bound the phase and code range rate	Related to GPS receiver's moving direction
Method 5	Doppler shift	Detect spoof that uses one transmitter to spoof all satellites	None
Method 6	Correlation peaks	Correlate L1/L2 binary message	Low performance on Y-code
Method 7	GPS signal after removing all navigation data	Recover authentic data	Requires low spoof/authentic signal power ratio
Method 8	Range differences: phase/code, L1/L2	Identify signal source	Needs to be L1/L2 receiver
Method 9	Ephemeris data	Verify ephemeris data including satellite position	None
Method 10	Signal power and data	Jump detection	None

not need to avoid cross-checking at the spoofed receiver, the position solution can be forged in advance and then the pseudo-range model used again to get a time sequence of the spoofed pseudo-range.

In one possible application of this sort of spoofing, consider spoofing of the vessel monitoring system (VMS). The VMS (typically employing GPS today) records the voyage of the vessel and automatically provides the data to the fisheries monitoring centre of the EU member state where the vessel is registered as well as the member state in whose waters the vessel is fishing. Naturally, the data can be used to detect passage into waters for which the vessel is not licensed.

In this case of spoofing, the intent of the operator would be to log a fictional voyage that does not disclose illegal fishing activities. In such a situation, all the spoofing has to do is to disconnect the GPS antenna and attach a local GPS signal generator instead.

Simple GPS fraud kits are available for about \$2535 that could feed spoofed signals into the RS-232 port

of the VMS after disabling the antenna and opening the VMS box. There are also sophisticated GPS signal simulators available for about \$126,700 that could be connected to or radiated towards the VMS unit.

Countermeasures to receiver-end spoofing

As a one-way broadcast system, GPS is not immune to spoof attack, except the Y-code whose encryption algorithm is unavailable to civilian users. A spoof can never be detected using check matrices, like the CRC check, in the digital domain. However, it's possible by cross-checking the observables, intermediate measurement and positioning solutions.

Various countermeasures to receiver-end spoofing are summarised in the table. We could also use angle-of-arrival discrimination. It uses a dual-antenna receiver based on observation of L1 carrier differences between multiple antennae referenced to a common oscillator. If the expected delta phase measurements do not agree with the expected phase profiles within bounds set by the expected noise and attitude

uncertainty, a spoofing signal is identified.

For a spoofer to defeat the algorithm as implemented, the spoofing system must emulate the expected carrier-phase deltas between the pair of antennae for all satellites in track. This geometry can't be emulated for several satellites if the spoofer is limited to one transmitting antenna.

A spoofer with two separate points of transmission might be able to defeat the algorithm. However, this would also require the spoofer to know the geometry of the GPS antenna array, locate a matched transmitter antenna very close to each GPS antenna, and deal with multipath, signal leakage and self-interference problems.

Why is spoofing the military GPS difficult?

While the GPS P-code is heavily encrypted and thus is hard to spoof, the civilian GPS signal (the C/A code), is easy to spoof because the signal structure, spread-spectrum codes and modulation methods are open to the public. This could make you think of a frightening scenario where terrorists mislead the GPS of GPS-guided missiles and bombs so that they hit an unintended target, thereby precipitating a crisis.

Manipulating the military GPS is, however, much more difficult. Considerable research has been carried out in this regard. Security design of the M-code signal is based on next-generation cryptography and other aspects, including a new keying architecture. The modulation of the M code signal is a binary offset carrier signal with subcarrier frequency of 10.23 MHz and spreading code rate of 5.115 MHz, denoted as BOC(10.23,5.115) or BOC(10,5) modulation; the spreading code transitions are aligned with square-wave subcarrier transitions. Spreading and data modulations employ biphasic modulation, so that the signal occupies one phase quadrature channel of the carrier. The spreading code is a pseudorandom bitstream from a signal